Decay of the Fourier Transform of Rvachev's Up-Function

Dyadic products, ergodic fluctuations, and sharp annular envelopes

A rigorous analysis for Vladimir Reshetnikov's `ProveIt` project

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Abstract

Let

$$f(x) = \widehat{\mathrm{up}}(x) = \prod_{n=0}^{\infty} \frac{\sin(\pi x/2^n)}{\pi x/2^n}$$

be the Fourier transform of Rvachev's compactly supported up-function, with the Fourier convention $\widehat{\mathrm{up}}(x) = \int_{\mathbb{R}} \mathrm{up}(t)\mathrm{e}^{-2\pi i x t}\,\mathrm{d}t$. The phrase "the decay rate of f" is intrinsically ambiguous: f vanishes at every nonzero integer, its values on a fixed dyadic ray are governed by a Birkhoff sum for the doubling map, and the largest lobe in a dyadic annulus is an extremal problem for a Thue–Morse trigonometric polynomial. These regimes have different second-order behavior.

Writing $a = \log 2$, we prove the exact dyadic decomposition that reduces the problem to the potential $\phi(t) = \log|\sin \pi t|$. For Lebesgue-almost every mantissa $y \in (1,2)$,

$$\log|f(2^N y)| = -\frac{\left(\log(2^N y)\right)^2}{2a} - \left(\frac{\log \pi}{a} + \frac{3}{2}\right)\log(2^N y) + o(N).$$

The centered remainder satisfies a central limit theorem with variance $\pi^2/4$ per dyadic step and a law of the iterated logarithm. Thus even the natural deterministic normalization does not converge to a nonzero constant. The slowest invariant average is attained by the period-two orbit $\{1/3, 2/3\}$, producing the power exponent

$$\kappa_* = \frac{\log \pi + a/2 - \log(\sqrt{3}/2)}{a} = \frac{\log \pi}{\log 2} + \frac{3}{2} - \frac{\log 3}{2\log 2}.$$

Finally, if $M_N = \max_{2^N \le x \le 2^{N+1}} |f(x)|$, then the classical Gelfond estimate and a boundary-layer optimization give

$$\begin{aligned}\log M_N = &-\frac{a}{2}N^2 + \left(\log(\sqrt{3}/2) - \log \pi - a/2\right)N \\ &- \frac{(\log N)^2}{2a} + \frac{\log N \,\log\log N}{a} + \mathcal{O}(\log N).\end{aligned}$$

Equivalently, the annular envelope has a genuine $-(\log\log x)^2/(2\log 2)$ correction, followed by a $(\log\log x)(\log\log\log x)/\log 2$ term. This rules out both a global pointwise equivalent and the fixed logarithmic-power correction proposed in the starting draft.

Contents

1 The question and the form of the answer

Rvachev's up-function up is an even, nonnegative C^∞ function supported on $[-1,1]$, normalized by $\int_{\mathbb{R}} \mathrm{up} = 1$. It is the density of an infinite sum of independent centered uniform random variables, and its Fourier transform is the infinite sinc product studied below; see [1, 5, 8]. Smooth compact support already implies that $f = \widehat{\mathrm{up}}$ decreases faster than every inverse power on the real axis. The product contains much more information: its dominant logarithmic scale is quadratic in $\log x$.

A preliminary draft in the `ProveIt` repository proposed a single pointwise equivalent of the form

$$|f(x)| \overset{?}{\sim} C(\log x)^{-p} x^{-\log(2\pi)/\log 2} \exp\left(-\frac{(\log x)^2}{2\log 2}\right). \tag{1}$$

That form cannot be correct as a pointwise statement. The first obstruction is immediate: $f(m) = 0$ for every nonzero integer m. The more substantive obstruction is dynamical. On writing $x = 2^N y$ with $1 \le y < 2$, the logarithm of the product contains

$$\sum_{k=1}^{N} \log|\sin(\pi 2^k y)|,$$

a lacunary Birkhoff sum. It has a typical mean, Gaussian-scale fluctuations, exceptional periodic averages, and an extremal average. No deterministic normalizer can erase all of these simultaneously.

There are therefore several mathematically natural questions.

1. What is the decay along a fixed dyadic ray $x_N = 2^N y$?
2. What happens for Lebesgue-typical mantissas y?
3. Which fixed rays decay most slowly?
4. How large can any lobe be in the dyadic annulus $[2^N, 2^{N+1}]$?

The article answers all four. The last question is the appropriate interpretation of a global upper envelope, while the second describes the amplitude seen at a typical point of a large dyadic annulus.

Main constants. Throughout, put

$$a = \log 2, \qquad \beta = \log\frac{\sqrt{3}}{2},$$

and define

$$\kappa_{\mathrm{typ}} = \frac{\log\pi + 3a/2}{a} = \frac{\log\pi}{\log 2} + \frac{3}{2} \approx 3.1514961295, \tag{2}$$

$$\kappa_* = \frac{\log\pi + a/2 - \beta}{a} = \frac{\log\pi}{\log 2} + \frac{3}{2} - \frac{\log 3}{2\log 2} \approx 2.3590148791. \tag{3}$$

The first is the typical power exponent. The second comes from the maximal Birkhoff average and controls the linear-in-$\log x$ term of the slow annular envelope.

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2 Product structure, zeros, signs, and lobes

We use the entire extension

$$f(z) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right), \qquad \operatorname{sinc} w = \begin{cases} \sin w/w, & w \ne 0, \\ 1, & w = 0. \end{cases} \tag{4}$$

The product converges locally uniformly because $\log\operatorname{sinc} w = -w^2/6 + \mathcal{O}(w^4)$ near the origin.

Proposition 2.1 (Equivalent products and dilation equations)**.** *For every $z \in \mathbb{C}$,*

$$f(z) = \prod_{m=1}^{\infty} \left(\cos \frac{\pi z}{2^m}\right)^m, \tag{5}$$

$$= \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}. \tag{6}$$

Moreover,

$$f(z) = \operatorname{sinc}(\pi z) f(z/2), \tag{7}$$

$$f(2z) = \operatorname{sinc}(2\pi z) f(z). \tag{8}$$

Proof. The identity $\operatorname{sinc} w = \prod_{j\ge 1} \cos(w/2^j)$ follows by iterating the double-angle formula and passing to the limit. In the product over $n \ge 0$, a factor $\cos(\pi z/2^m)$ occurs once for every decomposition $m = n + j$ with $n \ge 0$ and $j \ge 1$, hence exactly m times. This proves (5).

Euler's sine product gives

$$\operatorname{sinc}\left(\frac{\pi z}{2^n}\right) = \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{(2^n k)^2}\right).$$

A positive integer m can be written as $m = 2^n k$ for precisely $1 + \nu_2(m)$ values of $n \ge 0$. Regrouping the locally uniformly convergent product proves (6). Finally, separating the first factor in (4), or replacing z by $2z$, gives (7)–(8). □

Proposition 2.2 (Zeros and Thue–Morse signs)**.** *The nonzero zeros of f are exactly the integers. The zero at $m \in \mathbb{Z} \setminus \{0\}$ has multiplicity $1 + \nu_2(|m|)$. If $x > 0$ is not an integer and $n = \lfloor x \rfloor$, then*

$$\operatorname{sgn} f(x) = (-1)^{s_2(n)}, \tag{9}$$

where $s_2(n)$ is the sum of the binary digits of n.

Proof. The zero statement follows immediately from (6). Starting with $f(0) = 1$, the sign on $(n, n+1)$ is changed by the total multiplicity

$$\sum_{m=1}^{n} (1 + \nu_2(m)) = n + \nu_2(n!) = n + (n - s_2(n)) = 2n - s_2(n).$$

Its parity is the parity of $s_2(n)$, proving (9). □

The zeros divide the graph into unit lobes. Every lobe after the initial one has exactly one critical point; the initial lobe decreases from its endpoint maximum at the origin. This is useful when an "envelope" is interpreted as the sequence of lobe maxima.

Theorem 2.3 (Exactly one extremum in each noninitial unit interval)**.** *For each integer $n \ge 1$, the function $|f|$ has exactly one critical point in $(n, n+1)$. It is a strict maximum. Equivalently, the equation*

$$\sum_{m=1}^{\infty} \frac{1 + \nu_2(m)}{m^2 - x^2} = 0 \tag{10}$$

has exactly one solution in $(n, n + 1)$ for every $n \ge 1$.

Proof. Away from the integers, logarithmic differentiation of (6) gives

$$\frac{f'(x)}{f(x)} = -2xH(x), \qquad H(x) = \sum_{m=1}^{\infty} \frac{1+\nu_2(m)}{m^2-x^2}.$$

The series and its derivative converge locally uniformly off the integers, because $\nu_2(m) = \mathcal{O}(\log m)$. For $x > 0$,

$$H'(x) = 2x \sum_{m=1}^{\infty} \frac{1+\nu_2(m)}{(m^2-x^2)^2} > 0.$$

For $n \geq 1$, on $(n, n+1)$ one has $H(x) \to -\infty$ as $x \downarrow n$ and $H(x) \to +\infty$ as $x \uparrow n+1$. Hence H has exactly one zero there. At that zero the derivative of f'/f is negative, so $\log|f|$, and therefore $|f|$, has a strict local maximum. On $(0,1)$ every denominator in H is positive, so $f'/f < 0$ there; the initial lobe decreases strictly from $f(0) = 1$ to $f(1) = 0$. □

Remark 2.4. Let $A_n = \max_{n \leq x \leq n+1} |f(x)|$. Then the annular maximum introduced later is

$$M_N = \max_{2^N \leq n < 2^{N+1}} A_n.$$

Thus the annular theorem is also a theorem about the upper envelope of the unique extrema in successive lobes.

3 The exact dyadic decomposition

The decisive step is not an approximation. It is an exact decomposition into a deterministic quadratic term and a dynamical sum.

Let

$$T(t) = 2t \pmod 1, \qquad \phi(t) = \log|\sin \pi t|, \qquad \psi(t) = \log|2\sin \pi t| = \phi(t) + a. \tag{11}$$

Both potentials are understood as $-\infty$ at $t = 0$; they belong to $L^p(0,1)$ for every finite p.

Theorem 3.1 (Exact dyadic identity). *Let $N \geq 0$, $1 \leq y < 2$, and $z = y - 1$. Then*

$$\boxed{\log|f(2^N y)| = \log|f(y)| - N\log(\pi y) - \frac{aN(N+1)}{2} + \sum_{k=1}^{N} \phi(T^k z).} \tag{12}$$

The equality is interpreted as $-\infty = -\infty$ when an orbit point is zero.

Proof. Split the product (4) after the N factors whose arguments are at least of order one:

$$f(2^N y) = f(y) \prod_{k=1}^{N} \operatorname{sinc}(\pi 2^k y).$$

Because 2^k is an integer, $|\sin(\pi 2^k y)| = |\sin(\pi 2^k z)|$. Taking absolute values and logarithms gives (12), since $\sum_{k=1}^{N} k = N(N+1)/2$. □

Corollary 3.2 (Uniform log-normal upper scale). *As $x \to +\infty$,*

$$\log|f(x)| \leq -\frac{(\log x)^2}{2\log 2} + \mathcal{O}(\log x), \tag{13}$$

with an absolute implied constant. Consequently, for every $A > 0$, $f(x) = \mathcal{O}_A(x^{-A})$.

Proof. Write $x = 2^N y$ with $1 \le y < 2$ and use $|f(y)| \le 1$ and $\phi \le 0$ in (12). Since $N = (\log x - \log y)/a$, the quadratic term is $-(\log x)^2/(2a) + \mathcal{O}(\log x)$. The last assertion follows because a negative quadratic in $\log x$ eventually dominates $-A \log x$. $\square$

Warning 3.3. Equation (13) is a uniform upper bound, not a uniform equivalent. There can be exceptionally close returns of $T^k z$ to zero, and at dyadic rational points the product vanishes exactly. A matching lower bound cannot hold pointwise on the whole real axis.

4 Fixed dyadic rays and their Birkhoff averages

For $y \in (1, 2)$ put

$$z = y - 1, \qquad w = Tz,$$

so that

$$\sum_{k=1}^{N} \phi(T^k z) = \sum_{j=0}^{N-1} \phi(T^j w).$$

This shift makes the standard Birkhoff notation convenient.

Theorem 4.1 (Ray law from a finite Birkhoff average)**.** *Suppose $f(y) \neq 0$ and that the finite limit*

$$\alpha(y) = \lim_{N\to\infty} \frac{1}{N} \sum_{j=0}^{N-1} \phi(T^j w) \tag{14}$$

exists. For $x_N = 2^N y$,

$$\log |f(x_N)| = -\frac{(\log x_N)^2}{2a} - \kappa(\alpha(y)) \log x_N + o(\log x_N), \tag{15}$$

where

$$\kappa(\alpha) = \frac{\log \pi + a/2 - \alpha}{a}. \tag{16}$$

Proof. By (12),

$$\log |f(2^N y)| = -\frac{a}{2} N^2 + N\big(\alpha(y) - \log(\pi y) - a/2\big) + o(N).$$

Substitute $N = (\log x_N - \log y)/a$. The terms involving $\log y \cdot \log x_N$ cancel, leaving precisely (15)–(16). $\square$

The cancellation of $\log y$ is important: once the result is expressed in terms of the actual point x_N, the power exponent depends on the orbit average, not on the chosen representative of the mantissa.

4.1 The Lebesgue-typical law

The doubling map is ergodic for Lebesgue measure, and

$$\int_0^1 \phi(t)\, \mathrm{d}t = -\log 2 = -a. \tag{17}$$

The integral follows, for example, from the classical integral of $\log \sin$ on $(0, \pi)$.

Corollary 4.2 (Typical power exponent). *For Lebesgue-almost every $y \in (1,2)$,*

$$\boxed{\log|f(2^N y)| = -\frac{\left(\log(2^N y)\right)^2}{2\log 2} - \kappa_{\mathrm{typ}}\log(2^N y) + o(N),} \tag{18}$$

where κ_{typ} is given by (2). Equivalently,

$$|f(2^N y)| = \exp\left(-\frac{(\log(2^N y))^2}{2\log 2}\right)(2^N y)^{-\kappa_{\mathrm{typ}}+o(1)}. \tag{19}$$

Proof. Apply Birkhoff's ergodic theorem to $\phi \in L^1(0,1)$ and use (17) in equation (16). □

The power in the starting draft (1) was $\log(2\pi)/\log 2 = \log\pi/\log 2 + 1$. The typical exponent is larger by exactly $1/2$. More importantly, the error in (18) does not settle to a constant or even to a deterministic logarithmic correction; it has probabilistic fluctuations of order $\sqrt{N}$ and almost-sure iterated-logarithm excursions.

4.2 The maximal invariant average and a slow periodic ray

The same potential appears in the classical Thue–Morse trigonometric polynomial. Let

$$Q_N(t) = \prod_{j=0}^{N-1} 2|\sin(\pi 2^j t)|. \tag{20}$$

The factorization

$$\prod_{j=0}^{N-1}(1 - \mathrm{e}^{2\pi\mathrm{i}2^j t}) = \sum_{n=0}^{2^N-1}(-1)^{s_2(n)}\mathrm{e}^{2\pi\mathrm{i}nt} \tag{21}$$

shows that Q_N is the modulus of a Thue–Morse polynomial. The classical Gelfond estimate, and its ergodic-optimization interpretation, imply that there is a constant C_G such that

$$\sup_{t\in\mathbb{R}} Q_N(t) \le C_G(\sqrt{3})^N. \tag{22}$$

At $t = 1/3$ every factor has modulus $\sqrt{3}$, so the exponential rate is sharp. In the notation for ϕ, this says

$$\sup_t \sum_{j=0}^{N-1}\phi(T^j t) = N\beta + \mathcal{O}(1), \qquad \beta = \log\frac{\sqrt{3}}{2}. \tag{23}$$

The maximizing invariant measure is supported on the period-two orbit $\{1/3, 2/3\}$; see [3, 2, 9].

Proposition 4.3 (An exact slow ray). *Let $y = 4/3$ and $x_N = 2^N y$. Then*

$$|f(x_N)| = |f(4/3)|\left(\frac{3\sqrt{3}}{8\pi}\right)^N 2^{-N(N+1)/2}. \tag{24}$$

Equivalently,

$$|f(x_N)| = C_{4/3}\exp\left(-\frac{(\log x_N)^2}{2a}\right)x_N^{-\kappa_*}, \tag{25}$$

where

$$C_{4/3} = |f(4/3)|\exp\left(\frac{(\log(4/3))^2}{2a}\right)(4/3)^{\kappa_*} > 0. \tag{26}$$

Proof. For every $k \geq 1$, $|\sin(\pi 2^k \cdot 4/3)| = \sqrt{3}/2$. Substitute this into the exact finite product preceding (12). Rearranging with $\log x_N = aN + \log(4/3)$ gives (25). □

Remark 4.4. The adjective "slow" concerns the power correction after the universal quadratic term. It does not mean that (24) itself is the largest value in the annulus $[2^N, 2^{N+1}]$. The fixed mantissa $4/3$ pays an order-N penalty from y^{-N}. A larger lobe is obtained by choosing a mantissa much closer to 1 and allowing a transient of length about $\log_2(N/\log N)$ before the orbit enters $\{1/3, 2/3\}$. That boundary-layer optimization is the source of the double-log terms in section 6.

4.3 Why no global pointwise equivalent can exist

Proposition 4.5 (Failure of deterministic constant-amplitude normalization)**.** *There is no positive function $g(x)$, finite at all sufficiently large real x, for which*

$$g(x)|f(x)| \longrightarrow C \in (0, \infty) \tag{27}$$

as $x \to \infty$ through all real values. Even after restricting to Lebesgue-typical dyadic rays and using the natural centering from (18)*, the normalized amplitude has no almost-sure finite nonzero limit.*

Proof. The first statement follows from $f(m) = 0$ at every positive integer. The second follows from the fluctuation theorem in the next section: the centered logarithm has arbitrarily large positive and negative excursions on almost every ray. □

5 Gaussian fluctuations and the iterated logarithm

The typical law gives only the order-N mean. In this particular lacunary system, the next scale can be computed exactly.

Let P be the Perron–Frobenius operator of the doubling map,

$$(Ph)(x) = \frac{1}{2}\left(h(x/2) + h((x+1)/2)\right). \tag{28}$$

A direct trigonometric calculation gives

$$P\psi = \frac{1}{2}\psi. \tag{29}$$

Define

$$d = 2\psi - \psi \circ T. \tag{30}$$

Using $\sin(2u) = 2 \sin u \cos u$,

$$d(x) = \log|\tan \pi x|. \tag{31}$$

Moreover, $Pd = 0$, because the two inverse branches in (28) produce reciprocal tangent values. Thus $d \circ T^j$ is a stationary reverse martingale-difference sequence. From (30),

$$\sum_{j=0}^{N-1} \psi(T^j w) = \sum_{j=0}^{N-1} d(T^j w) - \psi(w) + \psi(T^N w). \tag{32}$$

The boundary term is negligible on the $\sqrt{N}$ scale.

Lemma 5.1 (Exact asymptotic variance)**.** *One has*

$$\int_0^1 d(x)^2 \, dx = \frac{\pi^2}{4}. \tag{33}$$

Proof. By (31) and symmetry,

$$\int_0^1 d(x)^2\,\mathrm{d}x = \frac{2}{\pi}\int_0^{\pi/2} \log^2(\tan u)\,\mathrm{d}u.$$

The standard beta-integral differentiation formulas give $\int_0^{\pi/2} \log^2(\tan u)\,\mathrm{d}u = \pi^3/8$, yielding (33). Equivalently, one may use the Fourier series $\psi(x) = -\sum_{m\geq 1}\cos(2\pi m x)/m$ and Parseval. □

It is useful to display the exact deterministic centering. For $y \in (1,2)$, put $b = \log y$ and

$$C_{\mathrm{typ}}(y) = \log|f(y)| + \frac{b^2}{2a} + \kappa_{\mathrm{typ}} b. \tag{34}$$

Then (12) is equivalent to

$$\log|f(2^N y)| + \frac{(\log(2^N y))^2}{2a} + \kappa_{\mathrm{typ}}\log(2^N y) - C_{\mathrm{typ}}(y) = \sum_{j=0}^{N-1}\psi(T^j w). \tag{35}$$

Theorem 5.2 (Central limit theorem)**.** *Choose y uniformly from $(1,2)$. Then, as $N\to\infty$,*

$$\frac{\log|f(2^N y)| + \frac{(\log(2^N y))^2}{2a} + \kappa_{\mathrm{typ}}\log(2^N y)}{(\pi/2)\sqrt{N}} \xrightarrow{\mathrm{d}} \mathcal{N}(0,1). \tag{36}$$

Proof. The variable $w = T(y-1)$ is Lebesgue distributed. Since $Pd = 0$, $d\circ T^j$ is a stationary reverse martingale-difference sequence. It has finite moments of every order and variance $\pi^2/4$. The martingale central limit theorem applies to the first sum on the right of (32); see, for example, [4]. The coboundary term divided by $\sqrt{N}$ tends to zero in probability. Finally, $C_{\mathrm{typ}}(y)/\sqrt{N} \to 0$ in probability; its logarithmic endpoint singularities are square-integrable. □

Theorem 5.3 (Law of the iterated logarithm)**.** *For Lebesgue-almost every fixed $y \in (1,2)$, let $R_N(y)$ denote the left-hand side of* (35)*. Then*

$$\limsup_{N\to\infty}\frac{R_N(y)}{(\pi/2)\sqrt{2N\log\log N}} = 1, \tag{37}$$

$$\liminf_{N\to\infty}\frac{R_N(y)}{(\pi/2)\sqrt{2N\log\log N}} = -1. \tag{38}$$

Proof. The martingale law of the iterated logarithm applies to $\sum_{j<N} d\circ T^j$, since d has moments of order $2+\delta$ for every $\delta > 0$, and the conditional quadratic variation has ergodic mean $\pi^2/4$; see [7]. The two coboundary terms in (32) are negligible on the $\sqrt{N\log\log N}$ scale. For example, the logarithmic singularity of ψ gives exponentially small tails, so Borel–Cantelli applies. □

Corollary 5.4 (Size of typical multiplicative oscillations)**.** *For almost every y, after removal of the quadratic and typical power factors, the remaining multiplier has subsequential logarithms of size*

$$\pm\frac{\pi}{2}\sqrt{2N\log\log N}. \tag{39}$$

Since $N \sim \log x/\log 2$, this is

$$\pm\frac{\pi}{2}\sqrt{\frac{2\log x}{\log 2}\log\log\log x} \quad \textit{on the logarithmic amplitude scale.} \tag{40}$$

In particular, a correction of order $p\log\log x$ cannot produce a pointwise constant normalized amplitude.

6 The sharp dyadic-annular envelope

Define

$$M_N = \max_{2^N \le x \le 2^{N+1}} |f(x)|. \tag{41}$$

This is the largest lobe amplitude in the Nth dyadic annulus. The exact identity becomes especially transparent after writing $x = 2^N(1+z)$, $0 \le z \le 1$:

$$\begin{aligned}\log|f(2^N(1+z))| = &-\frac{aN(N+1)}{2} - N\log\pi \\ &+ \Big[\log|f(1+z)| - N\log(1+z) + S_N(z)\Big],\end{aligned} \tag{42}$$

where

$$S_N(z) = \sum_{k=1}^{N} \phi(T^k z). \tag{43}$$

Let

$$B_N = \max_{0\le z\le 1} \{\log|f(1+z)| - N\log(1+z) + S_N(z)\}. \tag{44}$$

Then

$$\log M_N = -\frac{aN(N+1)}{2} - N\log\pi + B_N. \tag{45}$$

If one ignored the term $-N\log(1+z)$, the maximal Birkhoff sum would be $N\beta + \mathcal{O}(1)$, attained on the orbit $\{1/3, 2/3\}$. But that orbit has z bounded away from zero and therefore pays a linear penalty. The optimizer instead starts at a tiny preimage of $1/3$. A transient of length r costs about $ar^2/2$, while choosing $z \asymp 2^{-r}$ costs about $N2^{-r}$ through $-N\log(1+z)$. Balancing these losses gives $2^{-r} \asymp r/N$, hence

$$r = \frac{\log N - \log\log N}{\log 2} + \mathcal{O}(1). \tag{46}$$

We first isolate the elementary saddle estimate.

Lemma 6.1 (Boundary-layer saddle). *Fix $c > 0$ and $A \in \mathbb{R}$. As $N \to \infty$,*

$$\min_{r\in\mathbb{N}_0} \left\{\frac{a}{2}r^2 + cN2^{-r} + Ar\right\} = \frac{(\log N)^2}{2a} - \frac{\log N\ \log\log N}{a} + \mathcal{O}(\log N). \tag{47}$$

Proof. First omit Ar and minimize over real $r \ge 0$. With $q = ar$, the critical point satisfies

$$q\mathrm{e}^q = acN, \qquad q = W(acN),$$

where W is the principal Lambert function. At that point the minimum is

$$\frac{q^2 + 2q}{2a}.$$

The expansion $W(acN) = \log N - \log\log N + \mathcal{O}(1)$ gives the two displayed terms and an error $\mathcal{O}(\log N)$. Rounding r to an integer changes the value by $\mathcal{O}(\log N)$, because the second derivative at the saddle is $\mathcal{O}(\log N)$. Any minimizing integer is $\mathcal{O}(\log N)$; the added term Ar therefore also changes the result by only $\mathcal{O}(\log N)$. $\square$

Theorem 6.2 (Sharp dyadic-annular maximum). *As $N \to \infty$,*

$$\boxed{\begin{aligned}\log M_N = &-\frac{a}{2}N^2 + \big(\beta - \log\pi - a/2\big)N \\ &-\frac{(\log N)^2}{2a} + \frac{\log N\ \log\log N}{a} + \mathcal{O}(\log N).\end{aligned}} \tag{48}$$

Proof. We prove matching bounds for B_N.

Upper bound. The product representation gives $|f(1+z)| \le 1$. If $z > 1/4$, then $-N\log(1+z) \le -c_0 N$ for an absolute $c_0 > 0$, while (23) gives $S_N(z) \le N\beta + \mathcal{O}(1)$. This region is far below the eventual maximum, whose loss from $N\beta$ is only of order $(\log N)^2$.

Now suppose $0 < z \le 1/4$, and let r be the largest nonnegative integer such that

$$2^r z \le \frac{1}{4}. \tag{49}$$

If $r \ge N$, then all N factors lie in the small-angle regime and $S_N(z) \le -aN^2/2 + \mathcal{O}(N)$, again much too small. We may therefore assume $r < N$. For $1 \le k \le r$ there is no wrap modulo one, and $\sin(\pi 2^k z) \le \pi 2^k z$. Because $z \le 2^{-r-2}$,

$$\begin{aligned}\sum_{k=1}^{r} \phi(T^k z) &\le r\log\pi + r\log z + \frac{ar(r+1)}{2} \\ &\le -\frac{a}{2}r^2 + r\left(\log\pi - \frac{3a}{2}\right).\end{aligned} \tag{50}$$

The remaining $N-r$ terms satisfy the shifted Gelfond estimate

$$\sum_{k=r+1}^{N} \phi(T^k z) \le (N-r)\beta + \mathcal{O}(1).$$

Maximality in (49) also gives $z > 2^{-r-3}$. Since $\log(1+z) \ge z/2$ for $0 \le z \le 1$,

$$N\log(1+z) \ge 2^{-r-4}N.$$

Combining the last three estimates, for fixed constants $c, C > 0$,

$$\log|f(1+z)| - N\log(1+z) + S_N(z) \le N\beta - \frac{a}{2}r^2 - cN2^{-r} + Cr + C. \tag{51}$$

The saddle lemma implies

$$B_N \le N\beta - \frac{(\log N)^2}{2a} + \frac{\log N\ \log\log N}{a} + \mathcal{O}(\log N). \tag{52}$$

Lower bound. For an integer $1 \le r < N$, take the explicit preperiodic point

$$z_r = \frac{1}{3\,2^r}. \tag{53}$$

After r iterations it reaches $1/3$ and then alternates between $1/3$ and $2/3$. Therefore

$$S_N(z_r) = A_r + (N-r)\beta, \tag{54}$$

where

$$A_r = \sum_{j=0}^{r-1} \log\sin\left(\frac{\pi}{3\,2^j}\right). \tag{55}$$

Separating each small angle from its sinc correction gives

$$A_r = r\log(\pi/3) - \frac{ar(r-1)}{2} + \sum_{j=0}^{r-1} \log\operatorname{sinc}\left(\frac{\pi}{3\,2^j}\right)$$

$$= r\log(\pi/3) - \frac{ar(r-1)}{2} + \log f(1/3) + o(1). \tag{56}$$

The zero of f at 1 is simple, so

$$\log|f(1+z_r)| = \log z_r + \mathcal{O}(1) = -ar + \mathcal{O}(1). \tag{57}$$

Also, $\log(1+z_r) \le z_r$. Substitution in (44) yields

$$B_N \ge N\beta - \frac{a}{2}r^2 - \frac{N}{3\,2^r} + \mathcal{O}(r). \tag{58}$$

Choose r according to the saddle lemma. Then

$$B_N \ge N\beta - \frac{(\log N)^2}{2a} + \frac{\log N\ \log\log N}{a} + \mathcal{O}(\log N). \tag{59}$$

Together, (52) and (59) determine B_N to the required precision. Finally use (45). □

6.1 The envelope in the original variable

Let $X = 2^N$ and $L = \log X$. Since $N = L/a$, (48) becomes

$$\boxed{\begin{aligned}\log \max_{X\le x\le 2X} |f(x)| = &-\frac{L^2}{2a} - \kappa_* L \\ &-\frac{(\log L)^2}{2a} + \frac{\log L\ \log\log L}{a} + \mathcal{O}(\log L),\end{aligned}} \tag{60}$$

for dyadic $X \to \infty$. Equivalently,

$$\begin{aligned}\max_{X\le x\le 2X} |f(x)| = &\exp\left(-\frac{(\log X)^2}{2\log 2}\right) X^{-\kappa_*} \\ &\times \exp\left(-\frac{(\log\log X)^2}{2\log 2} + \frac{\log\log X\ \log\log\log X}{\log 2} + \mathcal{O}(\log\log X)\right).\end{aligned} \tag{61}$$

The final exponential factor is smaller than every fixed negative power of $\log X$, but larger than every fixed negative power of X. It is therefore a genuinely intermediate correction, invisible in an ansatz containing only $(\log X)^{-p}$.

Corollary 6.3 (Near-optimal boundary layer). *The explicit points used in the lower bound have*

$$x_{N,r} = 2^N\left(1 + \frac{1}{3\,2^r}\right), \qquad r = \log_2 N - \log_2 \log N + \mathcal{O}(1). \tag{62}$$

Consequently their relative displacement from the left endpoint is

$$\frac{x_{N,r} - 2^N}{2^N} = \Theta\left(\frac{\log N}{N}\right) = \Theta\left(\frac{\log\log X}{\log X}\right). \tag{63}$$

Proof. Equation (46) gives $2^{-r} = \Theta((\log N)/N)$, and (63) follows from (53). □

The theorem does not assert that every exact maximizer is one of the special preimages (53); rather, those points construct the correct lower asymptotic, while the upper proof shows that any competing point must pay the same boundary-layer cost to the displayed precision.

7 Comparison of the distinct decay regimes

The formulas are easiest to compare after extracting the universal factor

$$\mathcal{L}(x) = \exp\left(-\frac{(\log x)^2}{2\log 2}\right). \tag{64}$$

Regime	Power after $\mathcal{L}(x)$	Additional behavior
Lebesgue-typical dyadic ray	$x^{-\kappa_{\text{typ}}}$, $\kappa_{\text{typ}} \approx 3.151496$	Log-amplitude fluctuations have standard deviation $(\pi/2)\sqrt{\log_2 x}$ and obey an iterated-logarithm law.
Maximal invariant periodic ray	$x^{-\kappa_*}$, $\kappa_* \approx 2.359015$	The ray $x = (4/3)2^N$ has an exact constant multiplier after this normalization.
Largest lobe in $[X, 2X]$	$X^{-\kappa_*}$	Multiplied by the boundary-layer factor in (61), whose leading logarithm is $-(\log\log X)^2/(2\log 2)$.
Starting-draft ansatz	$x^{-\log(2\pi)/\log 2}$, exponent ≈ 2.651496	A fixed power of $\log x$ cannot match either typical fluctuations or the annular boundary layer.

Several conclusions follow.

1. The coefficient $1/(2\log 2)$ of $(\log x)^2$ is robust in all regular regimes because it comes from the deterministic denominator $\prod_{k\le N} 2^k$.
2. The coefficient of $\log x$ is not universal. It is the affine image (16) of a Birkhoff average.
3. At typical points, the next visible term is not a deterministic $\log\log x$ term. Random-looking lacunary fluctuations are much larger.
4. For the largest lobe, optimizing the time needed to enter the maximizing period-two orbit creates a quadratic function of $\log\log x$.
5. Because of exact integer zeros and arbitrarily deep near-zero returns, there is no nontrivial global lower envelope comparable to the upper one.

8 A general geometric ratio

The mechanism is not special to the notation of the up-function. For an integer $q \ge 2$, consider

$$f_q(x) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi x}{q^n}\right). \tag{65}$$

Writing $x = q^N y$ gives the exact analogue

$$\log|f_q(q^N y)| = \log|f_q(y)| - N\log(\pi y) - \frac{\log q}{2}N(N+1) + \sum_{k=1}^{N}\phi(T_q^k z), \tag{66}$$

where $T_q t = qt \pmod 1$ and z is the fractional part of y. Hence the universal quadratic coefficient becomes

$$-\frac{(\log x)^2}{2\log q}. \tag{67}$$

For Lebesgue-typical mantissas the average of ϕ is still $-\log 2$, so the typical power exponent is

$$\kappa_{\mathrm{typ}}(q) = \frac{\log \pi + (\log q)/2 + \log 2}{\log q}. \tag{68}$$

The slow envelope is controlled by the maximal invariant average of ϕ under T_q. For $q = 2$, the exceptional simplification is that the exact maximum is $\log(\sqrt{3}/2)$ and is realized by the period-two orbit $\{1/3, 2/3\}$, linking the problem directly to the classical Thue–Morse Gelfond exponent.

9 Conclusions

The Fourier transform of Rvachev's up-function has a clean leading scale but no single clean pointwise equivalent. The exact finite decomposition (12) is the organizing principle:

$$\text{quadratic denominator cost} \quad + \quad \text{doubling-map Birkhoff sum.}$$

It yields the following rigorous answer to the decay question.

- Uniformly, $|f(x)|$ is at most $\exp(-(\log x)^2/(2\log 2) + \mathcal{O}(\log x))$, hence it is super-polynomially small.
- On almost every fixed dyadic ray, the power exponent after this log-normal factor is κ_{typ}, but the logarithm retains Gaussian and iterated-logarithm fluctuations.
- The slowest invariant average is the Thue–Morse period-two average, giving exponent κ_*. It is realized exactly on rays such as $(4/3)2^N$.
- The largest lobe in a dyadic annulus has the sharper expansion (48)–(61); its correction is quadratic in $\log\log x$, not a fixed power of $\log x$.

Thus the proposed constant-amplitude normalization in the starting draft must be replaced by a regime-dependent statement. The typical law, exceptional periodic law, fluctuation theorem, and annular envelope together give a precise and internally consistent description of how fast the infinite sinc product decays.

A Auxiliary integral and covariance checks

For completeness, the Fourier series

$$\psi(x) = \log|2\sin \pi x| = -\sum_{m=1}^{\infty} \frac{\cos(2\pi m x)}{m} \tag{69}$$

holds in $L^2(0,1)$. Parseval gives

$$\int_0^1 \psi(x)^2 \,\mathrm{d}x = \frac{1}{2}\sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{\pi^2}{12}. \tag{70}$$

Because only frequencies divisible by 2^k survive when ψ is paired with $\psi \circ T^k$,

$$\int_0^1 \psi(x)\psi(T^k x)\,\mathrm{d}x = \frac{\pi^2}{12\, 2^k}. \tag{71}$$

Therefore the Green–Kubo variance is

$$\frac{\pi^2}{12} + 2\sum_{k=1}^{\infty} \frac{\pi^2}{12\, 2^k} = \frac{\pi^2}{4}, \tag{72}$$

in agreement with (33) and the martingale decomposition.

B A direct formula for the simple zero at one

From (6),

$$f(x) = (1 - x^2)G(x), \qquad G(x) = \prod_{m=2}^{\infty} \left(1 - \frac{x^2}{m^2}\right)^{1+\nu_2(m)}. \tag{73}$$

The product for G converges locally uniformly near $x = 1$ and $G(1) \neq 0$. Consequently,

$$f(1 + z) = -2G(1)z + \mathcal{O}(z^2), \tag{74}$$

which justifies (57) and quantifies the endpoint cost in the annular lower bound.

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